Ethical Aspects of Consulting — Written Reporting and Oral Presentation

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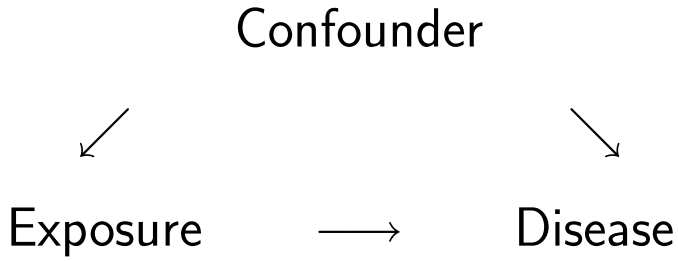
Part I

Ethical Behavior in a Consulting Context

Chapter 1
Some Case Studies

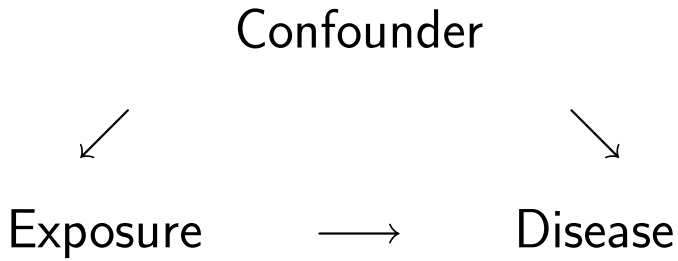
1.1 Statistics in the middle of a crisis: Environment and Health

- Waste incineration (dioxin exhaust) and congenital malformation

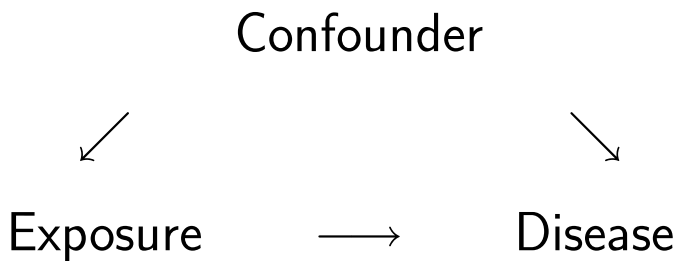


1.2 Statistics in the middle of a crisis: Smoking and lung cancer

- The tobacco industry versus the states of the United States
- Donald B. Rubin versus Scott L. Zeger



1.3 Statistics in the middle of a crisis: Cadmium in the north of Limburg



1.4 Paradigms

- Representation
- Comparison
  - ▷ Observational
  - ▷ Experimental (randomized)

1.5 Problems with Experiments: Psychology

! Not done: invention of studies/study results

? Not done: removal of study subjects that do not fit well with the rest

- $243 \times 3 = 729$
- SOLAS

1.6 Problems with Experiments: Randomized Clinical Trials

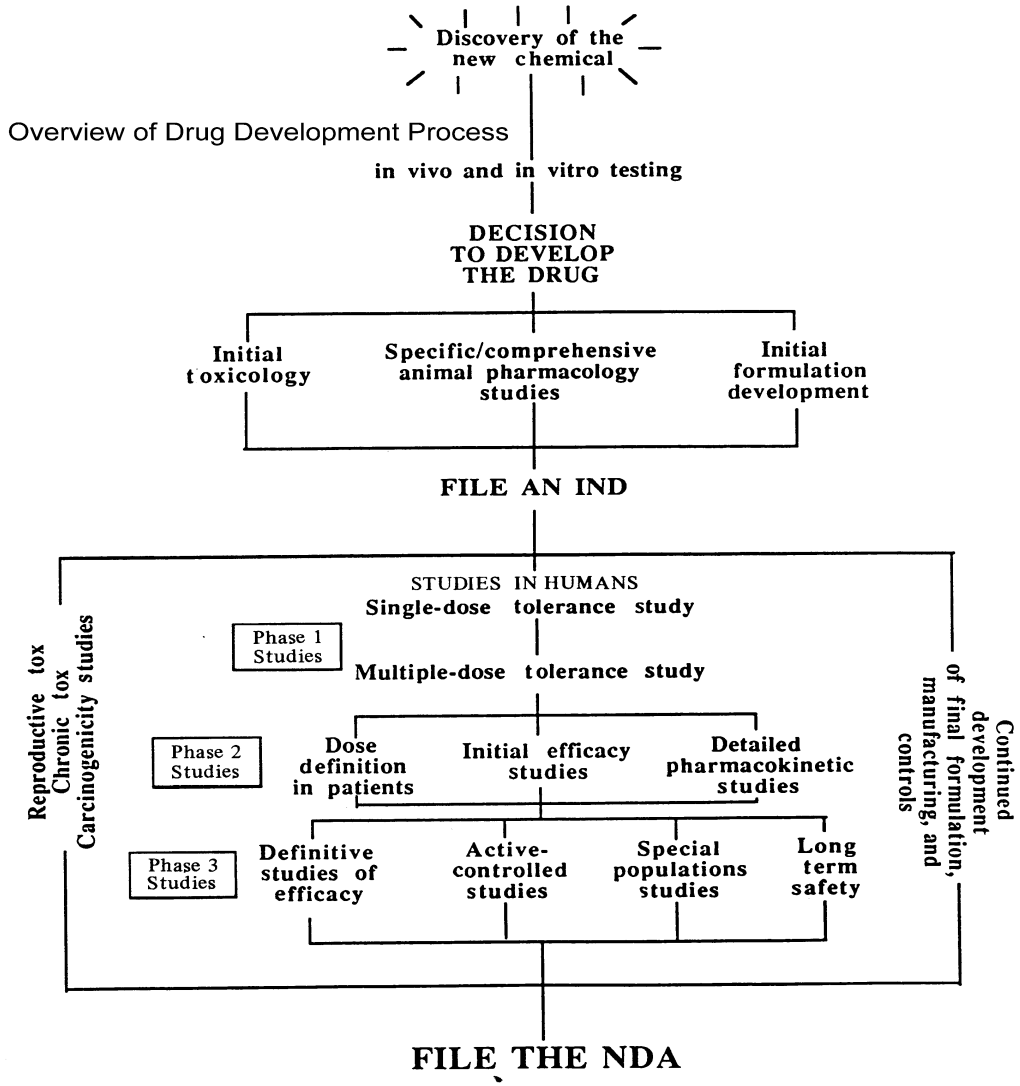
- ▷ The drug development process and regulatory issues
- ▷ Phases of clinical research
- ▷ Randomization
- ▷ Blindness
- ▷ Intention to Treat
- ▷ Ethical Considerations

1.7 Drug Development and Regulatory Issues

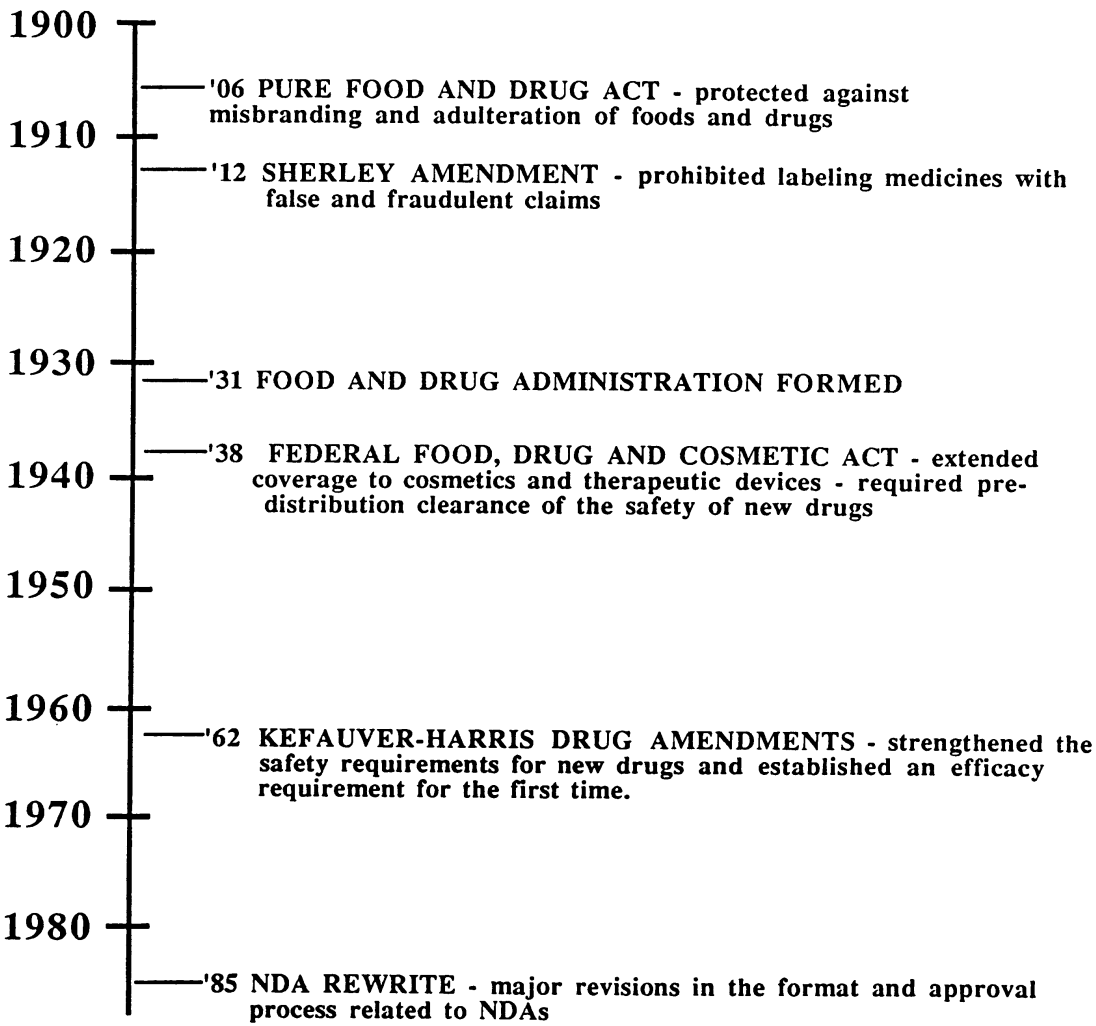
- Preclinical development:
  - ▷ Promising molecule discovered
  - ▷ Existing molecules help guide potential for activity
  - ▷ In vitro and in vivo testing
- Company management: Decision to further develop
- Several teams within company (sponsor) take over
- Decision that the drug has potential to be safe and effective

- IND is filed to FDA:
  - ▷ Description of the compound
  - ▷ All pre-clinical evidence obtained
  - ▷ Outline of first study in humans
- Conduct phase I–II–III studies
- File NDA

1.7.1 Drug Development Process



1.7.2 Food and Drug Administration



1.7.3 Notice of Claimed Investigational New Drug Exemption (IND)

- Originally designed to allow a non-approved drug to be shipped across state lines...
- Does it mean there is a legal vacuum within one state?
- No, states have their state-specific Regulatory Authorities
- Now taken to be the flexibility to test a drug in humans throughout the United States
- Framework within which drug development can take place
- Abbreviated by IND

1.7.4 New Drug Application (NDA)

- Request to FDA to market a new drug
- **All** evidence must be submitted (100s of volumes)
- Includes raw data
- Statistical analyses may be redone by the FDA
- They may decide to perform additional analyses
- **Label** is included
- High level of protection to the public
- Sufficiently fast and above all thorough review process

1.7.5 The European Context

- $27 + 1 = 28$
- Many countries: no statisticians at regulator
- Conflicts of interest
  - ▷ academic position
  - ▷ involvement in industry (as consultant)
  - ▷ involvement in regulator (as consultant)
- Last Observation Carried Forward

1.7.6 Drug Development and Statistics

- Pharmacokinetic and pharmacodynamic studies
- In vivo and in vitro studies
- Statistical design and analysis of phase I–II–III studies
- **Appropriate statistical section required in NDA**

1.7.7 Studies in Humans

- Required action/information:
  - ▷ Clinical study protocol
  - ▷ Investigator's education, training, experience
  - ▷ Information regarding hospital and other investigational institution
  - ▷ **Institutional Review Board:** independent scrutiny
  - ▷ Informed consent:
    - ∗ a written statement of the risks and benefits to the patient, prior to enrollment
    - ∗ signature of the patient that he/she has read and understood this information
  - ▷ **Independent Data Monitoring Committee**

1.8 Phases of Clinical Studies

1.8.1 Phase I

- First step to the human
- Very small sample size
- How good can the drug be tolerated: MTD
- MTD: maximal tolerated dose
- Methodologically rather under-developed!!!

1.8.2 Phase II

- Is the drug active?
- What is the rate of adverse events?
- Depends on phase I: if inadequate, the "wrong" dose is used
- Serves as input to phase III
- Methodological development quite limited

- Often staged or sequential trials are used:
  - ▷ A number of initial doses/initial drugs is considered
  - ▷ Non-promising doses are abandoned
  - ▷ Promising doses get more patients than initially planned

- **Differences with Phase III:**
  - ▷ Inclusion/exclusion criteria different
  - ▷ Different endpoint:
    - ∗ II: Tumor response versus III: Survival
    - ∗ II: CD4 count versus III: Survival

1.8.3 Phase III

- Designed to assess effectiveness of the new drug/intervention
- *The* clinical trial
- Since time frame in Phase III is relatively short, long term surveillance (also for side effects) is needed: Phase IV (these do not have control groups)
- Control groups: placebo or active (standard therapy)
- Randomization ensures groups are "alike", apart from intervention
  $\Longrightarrow$ causality

- Sufficient sample size needed
- Model for clinical practice !?
- Establishing adverse events
- *Cardiac Arrhytmia Suppression Trial (CAST 1989): commonly used antiarrhytmic drugs were harmful in patients who had had a myocardial infarction ! Encanaide and flecanaide They reduced arrhytmia but caused more than 3-fold increase in overall mortality !*

1.9 Randomization

Randomization tends to produce study groups comparable with respect to known and unknown risk factors, removes investigator bias in the allocation of participants, and guarantees that statistical tests will have valid significance levels.

\+ Comparable groups

– Representative for a general population

\+ Removal of investigator bias

\+ Statistical tests are valid

1.9.1 Methods of Randomization

- **Fixed allocation randomization**
  - ▷ Simple randomization
  - ▷ Blocked randomization
  - ▷ Stratified randomization
- **Adaptive randomization procedures**
  - ▷ Baseline adaptive randomization procedures
  - ▷ Responsive adaptive randomization

1.9.2 Simple Randomization

- Basic procedure
- Model: Drawing balls from urn
- Model: Tossing up a coin
- Easy to implement
- May result in imbalance, by pure chance
- One need to determine the proportion in each arm:
  - ▷ E.g., 1/2, 1/2
  - ▷ **1/3, 2/3**
  - ▷ E.g., 1/3, 1/3, 1/3
  - ▷ E.g., 1/4, 1/4, 1/2

1.9.3 Blocked Randomization

- Assign treatment in blocks (e.g., of size 4)
- Randomize patients within each block (e.g., on a 50–50 basis)
- Possibilities:

AABB	ABBA
ABAB	BABA
BAAB	BBAA

- Why randomization within a block?
- Imbalance is always restricted

1.9.4 Stratified Randomization

- Randomization $\Rightarrow$ groups are *on average* comparable
- In a particular study, imbalance could occur w.r.t., say, age or sex
- Solve this problem through stratified randomization:
  - ▷ Select a set of important baseline characteristics (age, sex,...)
  - ▷ Ensure these characteristics are measured
  - ▷ Determine groups within each characteristic:
    - ∗ Age: (1) 40–49 yr, (2) 50–59 yr, (3) 60–69 yr
    - ∗ Sex: (1) Male, (2) Female
    - ∗ Smoking history: (1) Current smoker, (2) Ex-smoker, (3) Never smoked

- ▷ The combination of stratifying variables defines *strata*
- ▷ In the example there are $3 \times 2 \times 3 = 18$ strata
- ▷ Perform randomization within strata

- The number of strata increases very quickly

- Adaptive randomization is an alternative, somewhat more complicated, technique to overcome baseline imbalance

1.10 Blindness

A clinical trial should, ideally, have a double-blind design to avoid potential problems of bias during data collection and assessment. In studies where such a design is impossible, a single-blind approach and other measures to reduce potential bias are favored.

- Unblinded trials
- Single-blind trials
- Double-blind trials
- (Triple-blind trials)

1.10.1 Unblinded Trial

- Synonym: Open trial
- Participant and investigator know intervention
- Examples:
  - ▷ surgical procedures
  - ▷ comparison of devices
  - ▷ comparison of treatment policy
  - ▷ changes in lifestyle

\+ Decision can be taken in a better informed manner

\+ Often, not always, simpler

– Bias

1.10.2 Single-blind Trials

- Participant is not aware of intervention
- Investigator knows intervention

\+ Patient is blinded $\Rightarrow$ less bias

\+ Investigator unblinded $\Rightarrow$ better safekeeping of patient's benefits

– But: bias is also likely, though maybe less pronounced than in unblinded trial

1.10.3 Double-blind Trials

- Participant and investigator are unaware of intervention
- Most prominent in drug efficacy studies

\+ Minimization of bias

– Logistics need to be done very carefully:

  ▷ Appropriate randomization method

  ▷ Certain actions cannot be done by the investigator and must be taken over by others

  ▷ Statisticians must remain unblinded as much as possible

  ▷ Data-monitoring committee

  ▷ Emergency unblinding procedure

1.10.4 Triple-blind Trials

- Participant and investigator are unaware of intervention
- *Also the* **Independent Data Monitoring Committee** *is unaware of treatment allocation*
- The DMC monitors endpoints in terms of A and B, without knowing allocation
- Since their role is to safeguard patients, this may lead to unease in investigators
- Especially when directionality matters (one does not want to continue a study until it is established that the new therapy is significantly worse...)

1.11 Intention to Treat

- Data may be incomplete:
  - ▷ Investigator can decide to take a patient off treatment
  - ▷ Informed consent: patients can withdraw at will
  - ▷ Patients can be lost to follow-up
- Whether or not such patients need to be included has to be decided
- There are two key principles:

- **Intention-to-treat principle** (Hill 1961):

> **An analysis that includes all randomized patients in the group to which they were randomly assigned, regardless of their adherence with the entry criteria, regardless of the treatment they actually received, and regardless of subsequent withdrawal from treatment or deviation from the protocol**

- **Treatment Received (As-treated):**

> **Patients are analyzed according to the treatment actually received, even if this is different from randomization.**

- Fierce debate continues.

1.12 Ethical Considerations

A properly designed and conducted clinical trial is ethical

+ Important public health questions are answered

– Patients may get an inferior treatment

- Can we continue a trial in which some trends already appeared $\longrightarrow$ early stopping: (group) sequential trials
- Public health perspective $\leftrightarrow$ benefit of an individual patient: which must prevail?
- International Ethical Guidelines for Biomedical Research Involving Human Subjects
- Informed consent is essential

- What about minors? Comatose patients?

- Randomization:
  - ▷ If physician believes there is a preferred treatment: the physician should perhaps not participate
  - ▷ Otherwise: randomization is not a problem, but a solution
  - ▷ Pro and con for a treatment may be studied at level of professional community: *clinical equipoise*

    The study may be started if there is sufficient uncertainty as to whether one of the two treatments is beneficial

- Patient should receive best standard care: placebo $\longrightarrow$ active control therapy

- Blinding is essential (but not always possible)

Chapter 2
Some Questions

- Genetically modified food/organisms: ethical or not?
- Screening programs: breast cancer, prostate cancer, cervix cancer, Down's syndrome,...
  - ▷ Sensitivity
  - ▷ Specificity
  - ▷ False positive
  - ▷ False negative
- 67,138 BEF

Chapter 3
The Toenail Data

- **T**oenail **D**ermatophyte **O**nychomycosis: Common toenail infection, difficult to treat, affecting more than 2% of population.

- Classical treatments with antifungal compounds need to be administered until the whole nail has grown out healthy.

- New compounds have been developed which reduce treatment to 3 months

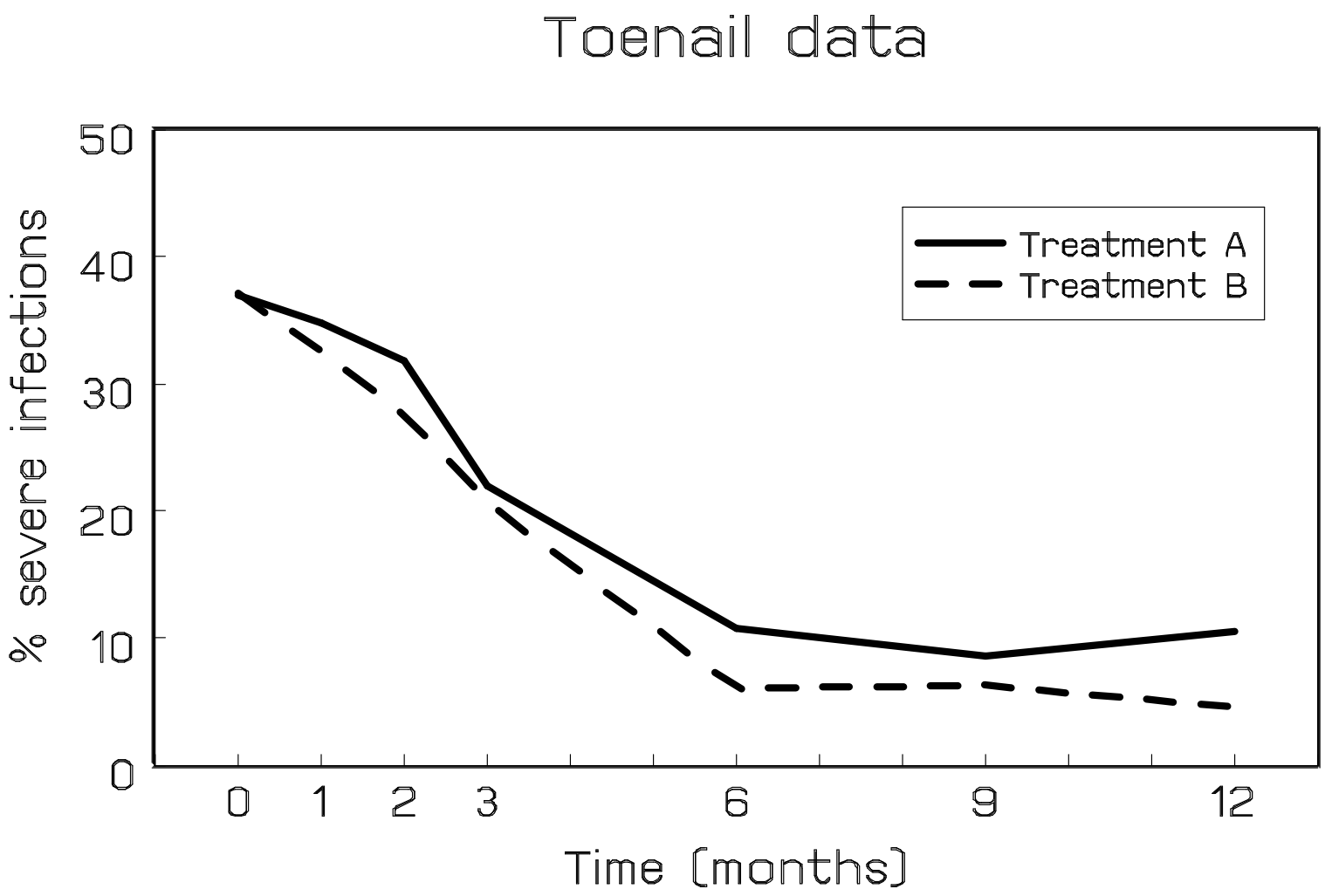
- Randomized, double-blind, parallel group, multicenter study for the comparison of two such new compounds (A and B) for oral treatment.

- Research question:

Severity relative to treatment of TDO ?

- 2×189 patients randomized, 36 centers
- 48 weeks of total follow up (12 months)
- 12 weeks of treatment (3 months)
- measurements at months 0, 1, 2, 3, 6, 9, 12.

- Frequencies at each visit (both treatments):



3.1 Application to the Toenail Data

- Consider the model:

$$\begin{aligned} Y_{ij} &\sim \text{Bernoulli}(\mu_{ij}) \\ \log\left(\frac{\mu_{ij}}{1-\mu_{ij}}\right) &= \beta_0 + \beta_1 T_i + \beta_2 t_{ij} + \beta_3 T_i t_{ij} \\ \text{Corr}(Y_{ij}, Y_{ij'}) &= \alpha \qquad \text{(working correlation)} \end{aligned}$$

- Y_{ij}: severe infection (yes/no) at occasion j for patient i
- t_{ij}: measurement time for occasion j
- T_i: treatment group

3.1.1 Standard GEE

- **SAS Code:**

```
proc genmod data=test descending;
class idnum timeclss;
model onyresp = treatn time treatn*time
                / dist=binomial;
repeated subject=idnum / withinsubject=timeclss
                         type=exch covb corrw modelse;
run;
```

- SAS statements:
  - ▷ The REPEATED statements defines the GEE character of the model.
  - ▷ 'type=': working correlation specification (UN, AR(1), EXCH, IND,...)
  - ▷ 'modelse': model-based s.e.'s on top of default empirically corrected s.e.'s
  - ▷ 'corrw': printout of working correlation matrix
  - ▷ 'withinsubject=': specification of the ordering within subjects

- **Selected output:**
  - ▷ Regression parameters:

Analysis Of Initial Parameter Estimates

Parameter	DF	Estimate	Standard Error	Wald 95% Confidence Limits		Chi-Square
Intercept	1	-0.5571	0.1090	-0.7708	-0.3433	26.10
treatn	1	0.0240	0.1565	-0.2827	0.3307	0.02
time	1	-0.1769	0.0246	-0.2251	-0.1288	51.91
treatn*time	1	-0.0783	0.0394	-0.1556	-0.0010	3.95
Scale	0	1.0000	0.0000	1.0000	1.0000	

  - ▷ Estimates from fitting the model, ignoring the correlation structure, i.e., from fitting a classical GLM to the data, using proc GENMOD.
  - ▷ The reported log-likelihood also corresponds to this model, and therefore should not be interpreted.
  - ▷ The reported estimates are used as starting values in the iterative estimation procedure for fitting the GEE's.

Analysis Of GEE Parameter Estimates
Empirical Standard Error Estimates

Parameter	Estimate	Standard Error	95% Confidence Limits		Z	Pr > |Z|
Intercept	-0.5840	0.1734	-0.9238	-0.2441	-3.37	0.0008
treatn	0.0120	0.2613	-0.5001	0.5241	0.05	0.9633
time	-0.1770	0.0311	-0.2380	-0.1161	-5.69	<.0001
treatn*time	-0.0886	0.0571	-0.2006	0.0233	-1.55	0.1208

Analysis Of GEE Parameter Estimates
Model-Based Standard Error Estimates

Parameter	Estimate	Standard Error	95% Confidence Limits		Z	Pr > |Z|
Intercept	-0.5840	0.1344	-0.8475	-0.3204	-4.34	<.0001
treatn	0.0120	0.1866	-0.3537	0.3777	0.06	0.9486
time	-0.1770	0.0209	-0.2180	-0.1361	-8.47	<.0001
treatn*time	-0.0886	0.0362	-0.1596	-0.0177	-2.45	0.0143

- The working correlation:

Exchangeable Working Correlation

Correlation 0.420259237

3.1.2 Inference on Key Parameter: β_3

Model	Estimate (s.e.)	p-value
Initial model	-0.0783 (0.0394)	0.0469
Model-based (naive)	-0.0886 (0.0362)	0.0143
Empirically corrected (robust)	-0.0886 (0.0571)	0.1208

3.1.3 Inference on Key Parameter: β_3. Story 1.

Model	Estimate (s.e.)	p-value
Initial model	-0.0783 (0.0394)	0.0469
Model-based (naive)	-0.0886 (0.0362)	0.0143
Empirically corrected (robust)	-0.0886 (0.0571)	0.1208

"The initial model is the most efficient estimator, because it assumes that each data point provides an independent piece of information. Based on this model, the treatment effect is marginally significant."

3.1.4 Inference on Key Parameter: β_3. Story 2.

Model	Estimate (s.e.)	p-value
Initial model	-0.0783 (0.0394)	0.0469
Model-based (naive)	-0.0886 (0.0362)	0.0143
Empirically corrected (robust)	-0.0886 (0.0571)	0.1208

"The model-based estimator assumes that the various pairs of measurements per patient exhibit a common correlation. This is estimated to be $\widehat{\alpha} = 0.42$*, considered to be a plausible value. Therefore, inferences are based on the model-based estimator; this leads to a significant effect of treatment, with* $p = 0.0143$*."*

3.1.5 Inference on Key Parameter: β_3. Story 3.

Model	Estimate (s.e.)	p-value
Initial model	-0.0783 (0.0394)	0.0469
Model-based (naive)	-0.0886 (0.0362)	0.0143
Empirically corrected (robust)	-0.0886 (0.0571)	0.1208

"The empirically-corrected estimator assumes that the various pairs of measurements per patient exhibit a common correlation, but that, at the same time, this correlation assumption may be incorrect. In other words, it protects against misspecification. Inferences are based on this estimator. We conclude that there is no significant effect of treatment, with $p = 0.1208$*."*

3.1.6 Inference on Key Parameter: β_3. Story 4.

Model	Working corr. α	Estimate (s.e.)	p-value
Initial model		-0.078 (0.039)	0.0469
Model-based (naive)	exchangeable	-0.089 (0.036)	0.0143
Emp. corr. (robust)	independence	-0.078 (0.055)	0.1515
Emp. corr. (robust)	exchangeable	-0.089 (0.057)	0.1208
Emp. corr. (robust)	unstructured	-0.114 (0.052)	0.0275

"The empirically-corrected estimator assumes that the various pairs of measurements per patient exhibit a certain structure, but that, at the same time, this correlation assumption may be incorrect. The working correlation that is closest to the true structure is generally most efficient. Inferences are based on this estimator, with unstructured working correlation. We conclude that there is a significant effect of treatment, with $p = 0.0275$*."*

3.2 The Generalized Estimating Equations Case: Discussion

- Nice method to efficiently and correctly analyze non-Gaussian longitudinal data

- But: there are pitfalls

  ▷ Initial $\longleftrightarrow$ Model-based $\longleftrightarrow$ Empirically corrected

  ▷ Various working correlation structures possible: **how** and **when** to choose?

- **Know the method! Know the pitfalls! Stand firm on principles!**

Chapter 4
Multiple Imputation

- Valid under MAR
- Useful with GEE, missing covariates, complex data structures,...
- Three steps:
  1. The missing values are **sampled** M times $\Longrightarrow$ M complete data sets
  2. The M complete data sets are analyzed by using standard procedures
  3. The results from the M analyses are combined into a single inference
- Rubin (1987), Rubin and Schenker (1986), Little and Rubin (1987)

4.1 Use of MI in Practice

- Many analyses of the same incomplete set of data
- A combination of missing outcomes and missing covariates
- As an alternative to WGEE: MI can be combined with classical GEE
- MI in SAS:

Imputation Task: PROC MI

↓

Analysis Task: PROC "MYFAVORITE"

↓

Inference Task: PROC MIANALYZE

4.2 Multiple Imputation: Ticket to Fraud?

- Code for imputations:

```
proc mi data=armd13 seed=486048 out=armd13a simple nimpute=10 round=0.1;
   var lesion diff4 diff12 diff24 diff52;
   by treat;
   run;
```

- ?

Part II

Reporting and Presentation

Chapter 5
Effective Communication: Preamble

Note: Pages 59–74 are based on course notes prepared by Geert Verbeke.

5.1 Is it n or $n-1$?

- The variance:

$$\widehat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (Y_i - \overline{Y})^2$$

- The variance:

$$\widehat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^{n} (Y_i - \overline{Y})^2$$

- Standard deviation $\longleftrightarrow$ standard error
- "Should I use mean or median?"
- "What is a confidence interval?"

Chapter 6
Use and Misuse of Statistics

- Errors in statistics
- Two types of errors
- Multiple testing
- Equivalence tests
- Significance versus relevance

6.1 Example: Captopril Data

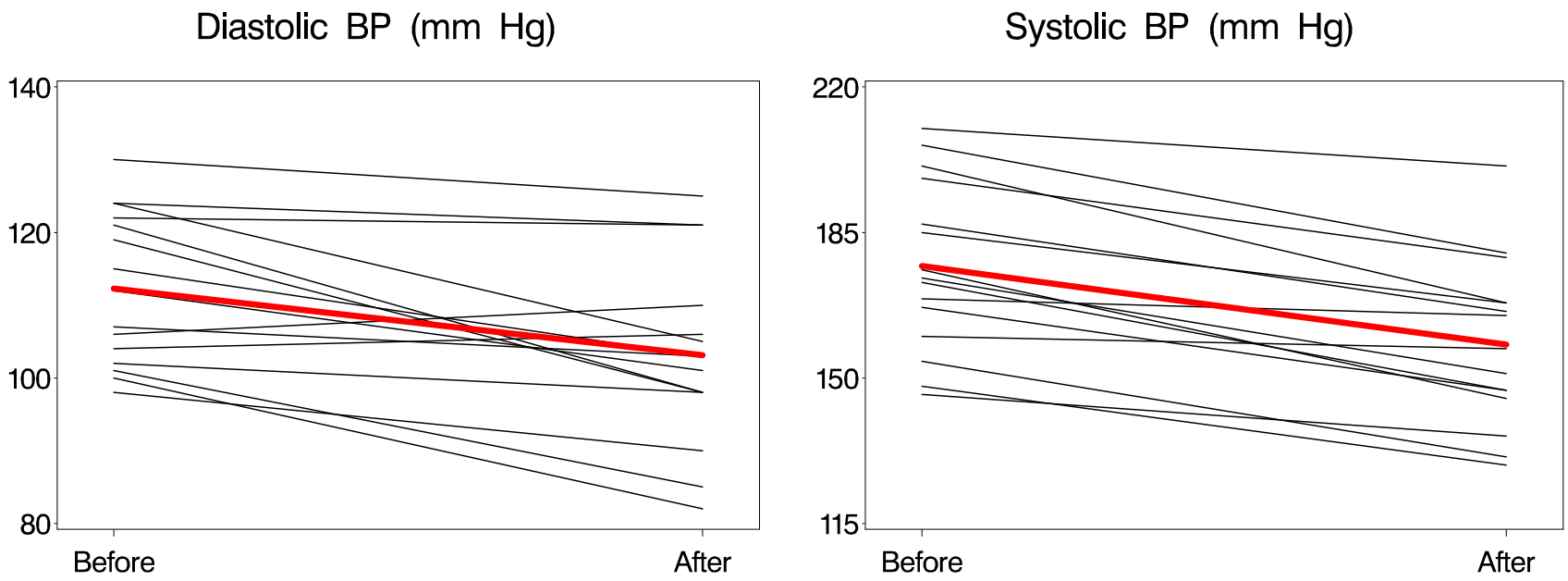
- 15 patients with hypertension
- The response of interest is the supine blood pressure, before and after treatment with CAPTOPRIL
- Research question:

How does treatment affect BP ?

- Dataset 'Captopril'

	Before		After	
Patiënt	SBP	DBP	SBP	DBP
1	210	130	201	125
2	169	122	165	121
3	187	124	166	121
4	160	104	157	106
5	167	112	147	101
6	176	101	145	85
7	185	121	168	98
8	206	124	180	105
9	173	115	147	103
10	146	102	136	98
11	174	98	151	90
12	201	119	168	98
13	198	106	179	110
14	148	107	129	103
15	154	100	131	82

	Average (mm Hg)
Diastolic before:	112.3
Diastolic after:	103.1
Systolic before:	176.9
Systolic after:	158.0



- "The result is 9.27 (4.91;13.63) with P =0.001."

- **"The difference in diastolic blood pressure is estimated as 9.27 mmHg, with 95% confidence limits [4.91,13.63] and p-value based on a two-sided t-test of 0.001."**

6.2 Possible Errors in Decision Making

- In our example about the Captopril treatment, we obtained $p = 0.001$ leading to the rejection of the null hypothesis of no treatment effect.
- This should not be considered as formal proof that there is a treatment effect
- Even if the treatment has no effect at all, a sample like ours would occur once every 1000 times.
- Maybe, our sample was indeed the extreme one that happens once every thousand experiments.
- Alternatively, suppose we would have obtained $p = 0.9812$. We then would not have rejected the null hypothesis, and concluded that there is no evidence for any treatment effect.

- This should not have been considered as formal proof that any treatment effect would be absent.

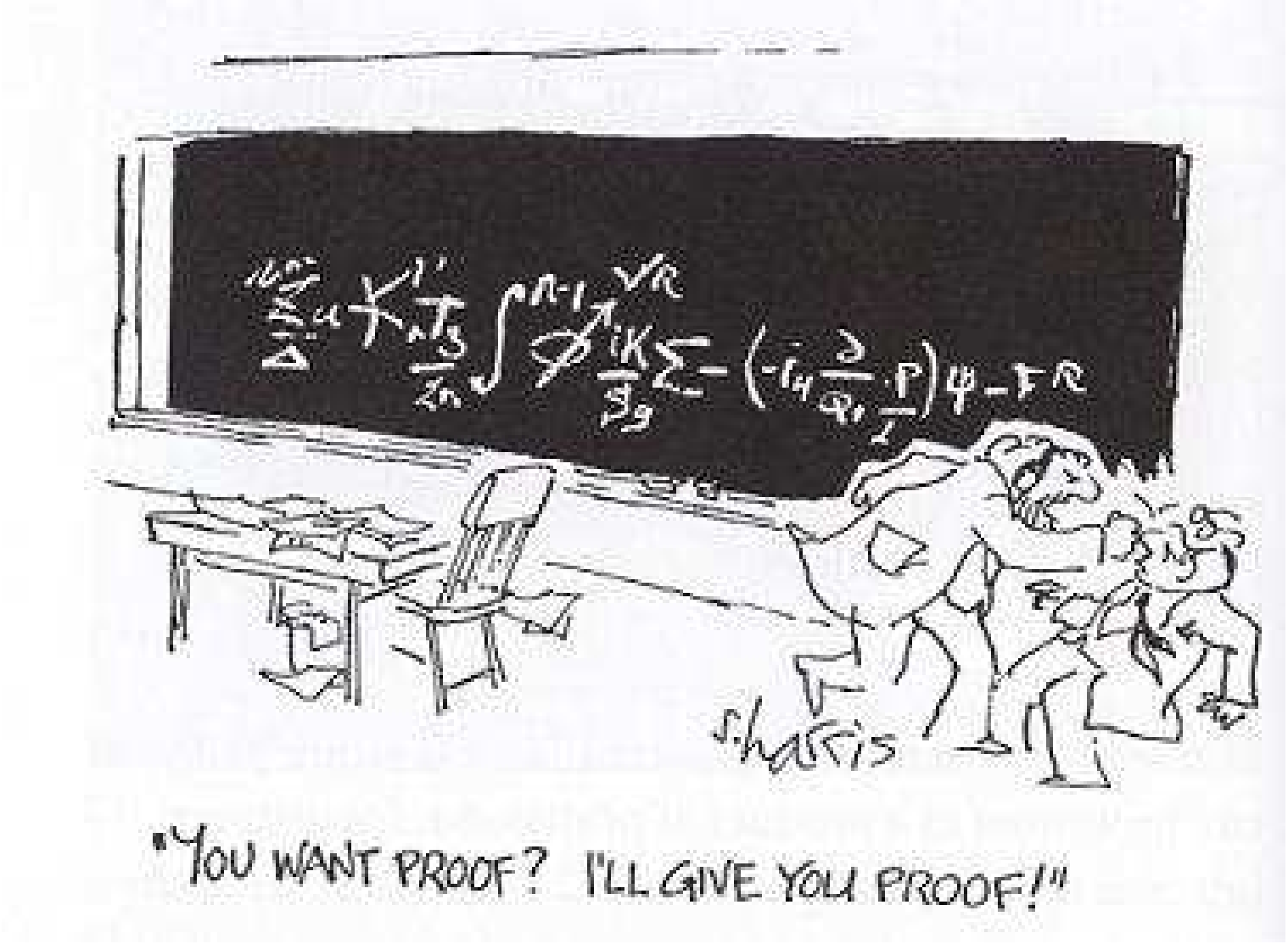
- Maybe, the treatment effect μ is not 0, but very close to 0. The data one then would observe would look very similar to data that would be observed if $\mu = 0$, such that the data do not allow to detect that $\mu \neq 0$

- Conclusion:

  ~~"Statistics can prove everything"~~

- **Intuitively:** Absolute certainty about population characteristics cannot be attained based on a finite sample of observations





"YOU WANT PROOF? I'LL GIVE YOU PROOF!"

6.3 Two Types of Errors

		Reality	
		H_0 correct	H_0 not correct
Test result	Accept H_0	**No error**	**Type II error**
	Reject H_0	**Type I error**	**No error**

- **Type I error:** H_0 is incorrectly rejected
- **Type II error:** H_0 is incorrectly accepted

6.4 Multiple Testing

- Each time a test is performed, there is probability α of making a type I error
- For example, if $\alpha = 0.05$, we can expect to incorrectly reject the null hypothesis in 5 out of 100 times.
- Implication:

> **"The more tests one performs, the higher the probability that something is detected by pure chance"**

- This problem of **multiple testing** occurs very frequently in bio-medical sciences, in various settings

6.4.1 Example: Subgroup analyses

- Kaplan et al., Table 5:

	All brain tumors		Benign		Malignant	
	Cases	Controls	Cases	Controls	Cases	Controls
Occupation	OR (CI)[a]		OR (CI)[a]		OR (CI)[a]	
Science professionals	3	6	2	3	1	3
	1.2	(0.3–5.1)	1.5	(0.2–9.8)	0.9	(0.1–10.0)
Engineers, architects, and technicians	4	14	2	5	2	9
	0.6	(0.2–2.0)	0.8	(0.1–4.5)	0.5	(0.1–2.7)
Health	3	9	2	8	1	1
	0.7	(0.2–2.8)	0.5	(0.1–2.5)	2.0	(0.1–39.3)
Executives and managers,	8	17	8	14	0	3
	1.1	(0.4–2.7)	1.2	(0.5–3.3)	Inf	
Teachers	6	15	4	10	2	5
	0.9	(0.3–2.6)	0.8	(0.2–3.0)	1.1	(0.2–6.9)
Office workers	26	65	16	41	10	24
	0.8	(0.5–1.3)	0.7	(0.4–1.5)	0.9	(0.4–2.1)
Telephone and radio operators, electricians	3	5	1	2	2	3
	1.2	(0.3–5.2)	1.0	(0.1–11.4)	1.4	(0.2–8.7)
Sales workers	15	24	8	14	7	10
	1.3	(0.6–2.5)	1.2	(0.5–3.0)	1.3	(0.4–3.7)
Housewife	34	60	24	39	10	21
	1.2	(0.7–2.0)	1.4	(0.7–2.8)	0.8	(0.3–2.3)
Domestic help	12	17	11	13	1	4
	1.4	(0.6–3.1)	1.8	(0.7–4.4)	0.4	(0.1–3.8)
Food	5	11	4	5	1	6
	0.8	(0.3–2.5)	1.5	(0.4–6.1)	0.3	(0.1–2.5)
Agricultural workers	2	13	0	9	2	4
	0.3	(0.1–1.3)			1.1	(0.2–6.1)
Metal workers	4	11	2	7	2	4
	0.7	(0.2–2.2)	0.5	(0.1–2.7)	0.9	(0.2–5.3)
Drivers, motor vehicle operators, and service occupations	17	15	8	10	9	5
	2.8	(1.3–6.2)	2.0	(0.7–6.1)	4.3	(1.3–14.2)
Construction workers	5	7	1	5	4	2
	1.3	(0.4–4.4)	0.3	(0.1–3.2)	3.9	(0.7–22.4)
Wood workers	3	3	2	1	1	2
	2.0	(0.4–10.1)	4.1	(0.4–48.0)	0.9	(0.1–11.4)
Weavers and tailors	11	8	7	4	4	4
	2.7	(1.1–7.1)	3.7	(1.1–13.5)	1.8	(0.4–8.0)
Printing workers	2	5	0	3	2	2
	0.7	(0.1–3.9)			2.0	(0.3–15.1)
Painters	2	3	0	1	2	2
	1.2	(0.2–7.2)			1.7	(0.2–12.9)
Unskilled workers	8	13	4	7	4	6
	1.2	(0.5–3.0)	1.1	(0.3–4.1)	1.4	(0.4–5.2)
Other	11	28	4	7	7	19
	0.8	(0.4–1.7)	1.2	(0.3–4.0)	0.6	(0.2–1.7)

*Adjusted for age, sex, and education.
[a]95% Confidence Interval in parenthesis.

- ▷ Tests based on C.I.'s for odds ratios
- ▷ C.I. containing 1 is equivalent to a non-significant test result
- ▷ $21 \times 3 = 63$ tests performed
- ▷ 5 significant results

6.4.2 Example: Searching for the Most Significant Results

- This 'scientific finding' was printed in the Belgian newspapers:



- It was even stated that those who wake up before 7.21am have a statistically significant higher stress level during the day than those who wake up after 7.21am.

6.4.3 Conclusion

- Significant results obtained by multiple testing are often overinterpreted
- If the number of tests is reported, the reader knows that such results need to be interpreted with extreme care
- The problem arises when only the significant results are reported, and one does not know how many tests were performed in total
- This leads to reporting results which turn out to be not reproducible
- For example, a new study would not find that students seated on the left are taller than those on the right. Instead, students seated on the left may weigh more than those seated on the right.

- For example, a new experiment might show no difference in stress levels between subjects waking up early and those waking up late. Or maybe a difference would be found only when waking up is later than 8.12am.

6.5 Equivalence Tests

- Suppose two groups A and B are to be compared, and a test is used to test

$$H_0 : \mu_A = \mu_B \quad \text{versus} \quad H_A : \mu_A \neq \mu_B$$

- In case of a non-significant test result, one often concludes that both groups are **identical** or **equivalent**

- An alternative interpretation is that the experiment did not have sufficient power to show an effect which is present.

- Conclusion:

> **Non-significance should not be interpreted as equivalence**

- This can also be seen from the fact that, if the test could be used to show equivalence, it would be best to collect data on (extremely) small samples, as this would increase the chance to obtain an non-significant result, due to lack of power.

- Instead, one should reverse H_0 and H_A:

$$H_0 : |\mu_A - \mu_B| > \Delta \quad \text{versus} \quad H_A : |\mu_A - \mu_B| \leq \Delta$$

  where Δ is a pre-specified constant, defining 'equivalence'

- Obviously, the result of the equivalence test entirely depends on the choice of Δ

- Therefore, Δ needs to be specified **prior to** the data collection

6.6 Significance versus Relevance

- We discussed before that the power to detect some effect Δ increases with the sample size

- This implies that any effect Δ, no matter how small, will, sooner or later, be detected, if the sample is sufficiently large.

- For example, consider the Captopril data, where the observed difference of 9.27 mmHg was found significantly different from zero ($p < 0.001$), based on data from 15 patients only:

Variable	Mean	Std.Dv.	N	Diff.	Std.Dv. Diff.	p
DIA_VOOR	112,3333	10,47219				
DIA_NA	103,0667	12,55540	15	9,266667	8,614495	0,000951

- Suppose that the observed difference would have been 0.1 mmHg.
- A p-value as small as 0.001 would be likely to be obtained, provided that the sample would be sufficiently large.
- Obviously, an average change in BP as small as 0.1 mmHg is not relevant from a clinical point of view.
- Conclusion:

Statistical significance $\neq$ **Clinical relevance**

- The p-value cannot distinguish between both situations
- It is therefore important not to blindly overinterpret significant results without knowing the size of the effect

Chapter 7
Written Reports

7.1 Preamble

- Who do I write for? "Who do I talk to?"
- Your audience may **not** be interested in statistics!
- Quotation and plagiarism
- Nobody has time to read a lot!
- Peculiarities of the English language:
  - ▷ color $\longleftrightarrow$ colour
  - ▷ analyze $\longleftrightarrow$ analyse
  - ▷ the data are ready $\longleftrightarrow$ the data is ready

7.2 Structure

- No absolute guideline
- Some useful models
  - ▷ For scientific articles in the substantive literature
  - ▷ For scientific articles in the (applied) statistical literature
  - ▷ For reports with statistical contents

- A possible structure (this is not the law!):
  - ▷ Title page with title, author(s), affiliations, date,...
  - ▷ Abstract/summary
  - ▷ Introduction (focus on application, then also a bit on statistical methodology)
  - ▷ Research question/data
  - ▷ Methodology
  - ▷ Analysis results
  - ▷ Discussion
  - ▷ Conclusion

7.3 Citation of literature

Don't: "As said by Professor Geert..."

Don't: "As stated by Professor Molenberghs..."

Don't: "As said by Geert M..."

Do: "As stated by Molenberghs (2005)..."

7.4 List of References

- Styles:
  - ▷ Harvard style
    - * http://en.wikipedia.org/wiki/Harvard_referencing
    - * http://guides.is.uwa.edu.au/harvard
    - * http://libweb.anglia.ac.uk/referencing/files/ShortHarvardGuide2012Sept.pdf
  - ▷ Vancouver style
    - * http://en.wikipedia.org/wiki/Vancouver_system
    - * http://www.library.uq.edu.au/training/citation/vancouv.pdf
  - ▷ ...
- Consistency!

7.5 List of References

Bahadur, R.R. (1961) A representation of the joint distribution of responses of n dichotomous items. In: *Studies in Item Analysis and Prediction*, H. Solomon (Ed.), Stanford Mathematical Studies in the Social Sciences VI. Stanford, California, Stanford University Press.

Liang, K.-Y. and Zeger, S.L. (1986). Longitudinal data analysis using generalized linear models. *Biometrika* **73**, 13–22.

Parzen, M., Lipsitz, S.R., Fitzmaurice, G.M., Ibrahim, J.G., and Troxel, A. (2006). Pseudo-likelihood methods for longitudinal binary data with non-ignorable missing responses and covariates. *Statistics in Medicine* **25**, 2784–2796.

Rubin, D.B. (1976). Inference and missing data. *Biometrika* **63**, 581–592.

Scharfstein, D.O., Rotnitzky, A., and Robins, J.M. (1999). Adjusting for nonignorable drop-out using semi-parametric nonresponse models. *Journal of the American Statistical Association* **94**, 1096–1120.

Van der Laan, M.J. and Robins, J.M. (2003). *Unified Methods for Censored Longitudinal Data and Causality*. Springer, New York.

Verbeke, G. and Molenberghs, G. (2000). *Linear Mixed Models for Longitudinal Data*. Springer, New York.

Chapter 8
Oral Presentation

8.1 The data on ‘data’ is and are mixed

The leading article in the November edition, arguing that the word data should be singular, attracted more letters from readers than any other topic in recent times. For the final word we shall publish on the matter, we turn to someone for whom English is not his native tongue, former Applied Statistics and current Biometrics editor Geert Molenberghs.

I read Ian Schagens and the ensuing correspondence with great interest since I, undoubtedly like many, have often pondered over the number of the noun data. Some of the arguments brought forward are interesting, logical and compelling. For example, a divorce between a Latin singular and plural form of a noun, coupled with diverging evolutions in their new home language, be it English or another living tongue, is quite common. Next to agenda, quoted by Schagen, another common example is medium and media, the latter commonly referring to mass media.

However, the English language being its beautiful but treacherous quicksand self, I got a wee bit suspicious when the original authors arguments got too logical, in the sense that rules of the English language, apparently above the slightest ground of doubt, were quoted, together with logic about structure in the physical data, or lack thereof, as to why data ought to be used singularly.

Since English, unlike, for example, French and Spanish, is not governed by a single authoritative body, we have the luxury of consulting several sources and, sure enough, we will stumble upon mixed pieces of advice.

In the Oxford English Grammar, Section 4.7 on irregular plurals reserves a lemma to the use of foreign plurals. It is stated that “Certain nouns in a are regularly treated as singular, though the ending represents an original plural: agenda, insignia. The use of other nouns in a as singulars is controversial. They include criteria, media, phenomena, strata. . . . Data is commonly used as a non-count noun in scientific discourse.” The Oxford English Grammar goes on to give a few examples (the first one abridged):

- “In this context inertia is. . . a combination of how much data weve got and then how far away it is from the average.”
- “Regression is commonly used by statisticians to calculate the best-fit through a set of data points in order to establish how close the data is to the ideal.”

While this apparently supports Schagens position, there is some hesitation in the sense that data is placed in the controversial category. It is therefore

useful to travel westward across the Atlantic and consult Websters New World English grammatical rulings

The American authority states (p14): "Other nouns are plural in form but may be either singular or plural, depending by the speaker or writer: ...

- Ethics is the study of a system of morals.
- The mans ethics are certainly questionable at best.

... Other words in this category include the following: data, strata, phenomena, statistics ...
Some nouns may be either singular or plural, depending on whether the context refers to a group or the individuals that comprise the group. These words ... are sometimes a bit confusing to work with, because a writers intention may not be understood, and a writer cannot be certain how a reader will interpret his or her written statements."

This is clearly relevant for our data case. If one, alongside Schagen, wants to emphasise the unity of data, let it be singular. However, at odds with Schagens view, one can also argue for the reverse, by stressing the collection of data points; then plural it shall be. This point of view was adopted by quite a few correspondents, too.

For this and other words, Webster concludes with a useful piece of advice—"How can you be certain whether the form of a noun is singular, plural, or both? The answer is simple: Consult a good dictionary."—were it not for the lack of unanimity amongst the said good English dictionaries. As a case in point, the Shorter Oxford English Dictionary asserts that data is "plural & collective singular," whereas Websters Third New International Dictionary simply states that data is plural of datum."

What do grammatical authorities recommend to professional and scientific writers? Stilman, a North American grammarian, (Grammatically Correct p.221), features the following example sentence: "We wont know the outcome until our data have been fully analyzed." She offers the following clarification: "Data is the plural of the rarely (if ever) used datum, meaning a single piece of factual information, such as a measurement or a statistic. As a plural noun, it properly takes a plural verb. However, it is increasingly being treated as a singular noun meaning the total body of facts accumulated. Which way you handle it may depend on the expectations of your audience. In scientific and academic writing, plural usage is the only correct way; in linguistically laxer environments, such as corporations, singular usage is accepted and may even be required. (It must be acknowledged that words do evolve. For example, the uncontroversial singular agenda, a list of individual to-do items, originated as a plural of agendum.)"

Thus, there is quite a bit of controversy, I daresay even confusion. Unlike Schagens suggestion, these are not confined to nor do they originate from the statistical community. Rather, the grammarians, in par with the history of the English language, take different perspectives. In the British Isles, there is

singular preference with plural tolerance, with the reverse holding at the other side of the North Atlantic. We should therefore perhaps abandon the rationally seductive but linguistically suicidal mission of coercing all use of data into singularity.

As an aside, the argument that the use of data as plural is inspired by people who want to display their acquaintance with Latin and therefore to be rejected is rather interesting, given a sizeable portion of the English vocabulary is a living testimony to yesteryears fashion of showing off ones command of French. To add to the relativity of it all, it is insightful to dwell on other living languages' customs of dealing with words of Latin origin. In Dutch, for example, datum is used for a calendar date, taking the plural form data. At the same time, data is employed, always in plural form, as the non-count term for a collection of data items, even though purists would claim it is better to use the 'proper' Dutch gegevens, plural of gegeven.

You may have noted that I glossed over Websters listing of statistics as a member of the dual plural-singular nouns category, the number being determined by the context. 'The statistics on British use and American usage or even utilization of data reveals mixed behavior, testifying that statistics remains a rich and diverse profession.'

In conclusion, we should refrain from adopting too rigid an attitude. As is clear from the Stilman citation, a language, especially English, is a living, organic creature. As one author observed: "Our language is ever adapting itself to changing circumstances. It is slowly shifting from day to day. The vigilant observer can detect its drift and slope. Now and then he discerns trends and tendencies which seem to him to adumbrate or prefigure the modifications of the next few centuries." We should not be afraid of partaking in this endeavour, since the same author asserted: "As in the past, so in the future, it will shape and adapt itself unceasingly to meet new needs, and in that incessant reshaping and adaptation every speaker and writer, consciously or unconsciously, will play some part." This author was Simeon Potter in his 1950 treatise Our Language (Harmondsworth: Penguin Books Ltd.). In this spirit, I read Schagens contribution, together with many, as an opportunity to carefully reflect on the way we handle data, statistically of course, but simultaneously from a linguistic standpoint.

8.2 The Data on 'Data' Is and Are Mixed

- The leading article in the November edition, arguing that the word data should be singular, attracted more letters from readers than any other topic in recent times.

- For the final word we shall publish on the matter, we turn to someone for whom English is not his native tongue, former Applied Statistics and current Biometrics editor Geert Molenberghs.

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another living tongue, is quite common.

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- It is stated that "Certain nouns in a are regularly treated as singular, though the ending represents an original plural: agenda, insignia. The use of other nouns in a as singulars is controversial.

- They include criteria, media, phenomena, strata. ... Data is commonly used as a non-count noun in scientific discourse."

- The Oxford English Grammar goes on to give a few examples (the first one abridged):
  - ▷ "In this context inertia is...a combination of how much data weve got and then how far away it is from the average."
  - ▷ "Regression is commonly used by statisticians to calculate the best-fit through a set of data points in order to establish how close the data is to the ideal."

- ...

8.3 The Data on 'Data' Is and Are Mixed

- **English** $\longleftrightarrow$ **French, Spanish**
- 'The data is...' $\longleftrightarrow$ 'The data are...'

	Oxford	**Webster**
Grammar	singular	both
Dictionary	$\pm$plural	plural

- datum $\longrightarrow$ data
- statistic $\longrightarrow$ statistics
- agendum $\longrightarrow$ agenda
- **Key message: be consistent within one text...**

8.4 The Toenail Data

The data were obtained from a randomized, double-blind, parallel group, multi-center study for the comparison of two oral treatments (in the sequel coded as A and B) for toenail dermatophyte onychomycosis (TDO), described in full detail by De Backer et al. (1996). TDO is a common toenail infection, difficult to treat, affecting more than 2 out of 100 persons (Roberts 1992). Anti-fungal compounds, classically used for treatment of TDO, need to be taken until the whole nail has grown out healthy. The development of new such compounds, however, has reduced the treatment duration to 3 months. The aim of the present study was to compare the efficacy and safety of 12 weeks of continuous therapy with treatment A or with treatment B.

In total, 2×189 patients, distributed over 36 centers, were randomized. Subjects were followed during 12 weeks (3 months) of treatment and followed further, up to a total of 48 weeks (12 months). Measurements were taken at baseline, every month during treatment, and every 3 months afterwards, resulting in a maximum of 7 measurements per subject. At the first occasion, the treating physician indicates one of the affected toenails as the target nail, the nail which will be followed over time. We will restrict our analyses to only those patients for which the target nail was one of the two big toenails. This reduces our sample under consideration to 146 and 148 subjects, in group A and group B, respectively.

One of the responses of interest was the unaffected nail length, measured from the nail bed to the infected part of the nail, which is always at the free end of the nail, expressed in mm. This outcome has been studied extensively in Verbeke and Molenberghs (2000).

Owing to a variety of unknown reasons, the outcome has been measured at all 7 scheduled time points for only 224 (76%) out of the 294 participants. Table 8.1 summarizes the number of repeated measurements available per subject,

Table 8.1: *Toenail Data. Number of available repeated measurements per subject, for each treatment arm.*

	Group A		Group B	
# Obs.	N	%	N	%
1	4	2.74%	1	0.68%
2	2	1.37%	1	0.68%
3	4	2.74%	3	2.03%
4	2	1.37%	4	2.70%
5	2	1.37%	8	5.41%
6	25	17.12%	14	9.46%
7	107	73.29%	117	79.05%
Total:	146	100%	148	100%

per treatment group. We observe that the occurrence of missingness is similar in both treatment groups for longer, but not for shorter sequences.

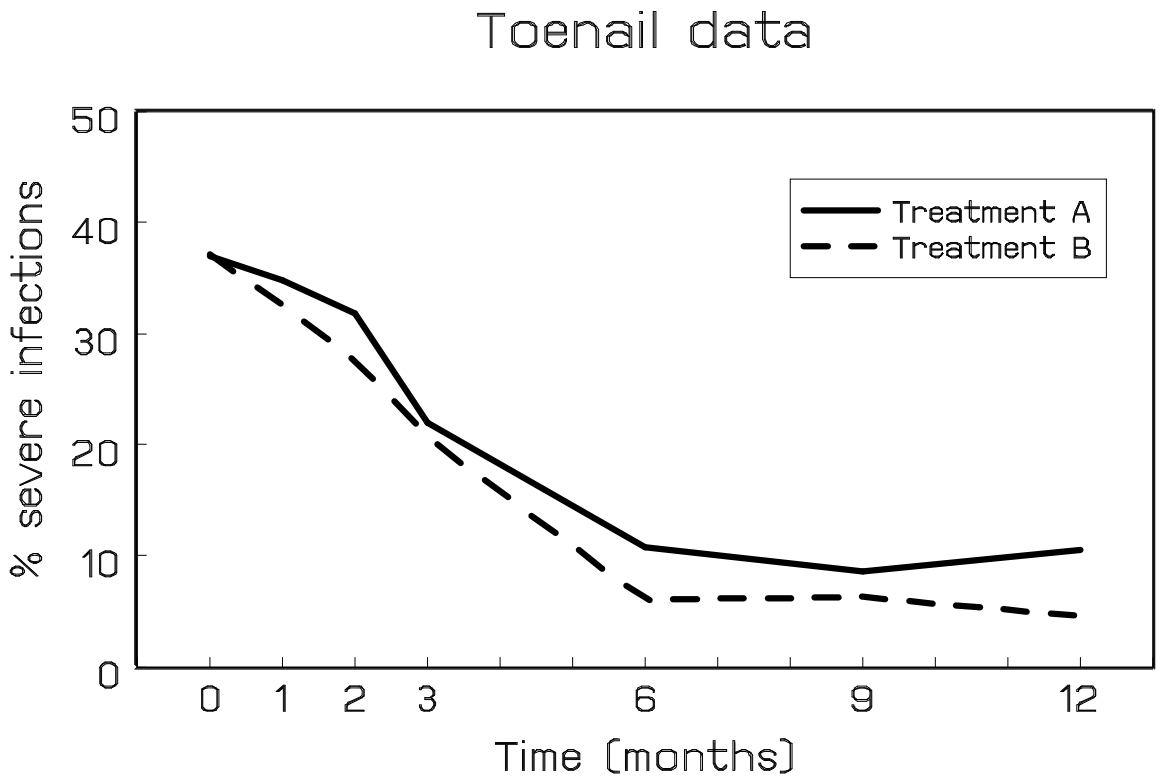
8.5 The Toenail Data

- **T**oenail **D**ermatophyte **O**nychomycosis: Common toenail infection, difficult to treat, affecting more than 2% of population.
- Classical treatments with antifungal compounds need to be administered until the whole nail has grown out healthy.
- New compounds have been developed which reduce treatment to 3 months
- Randomized, double-blind, parallel group, multicenter study for the comparison of two such new compounds (A and B) for oral treatment.
- Research question:

> **Severity relative to treatment of TDO ?**

- 2×189 patients randomized, 36 centers

- 48 weeks of total follow up (12 months)
- 12 weeks of treatment (3 months)
- measurements at months 0, 1, 2, 3, 6, 9, 12.
- Frequencies at each visit (both treatments):



8.6 Epilepsy Data: Posterior summary statistics for the the additive and multiplicative models.

Par.	Mean		SD		MC error		95% Credible interval		$\widehat{R}$	
	Add	Mult	Add	Mult	Add	Mult	Add	Mult	Add	Mult
β_{00}	0.4696	0.6485	0.1874	0.1854	0.0048	0.0042	$(0.0895, 0.8379)$	$(0.2813, 0.9949)$	1.00	1.00
β_{01}	0.736	0.9117	0.1808	0.1835	0.0045	0.0042	(0.3755,1.094)	(0.5538,1.269)	1.00	1.00
β_{10}	-0.0133	-0.0117	0.0076	0.0075	1.42E-04	1.52E-04	(-0.0284, 0.0012)	(-0.0262, 0.0036)	1.00	1.01
β_{11}	-0.0273	-0.0249	0.0078	0.0078	1.38E-04	1.53E-04	(-0.0430, -0.0119)	(-0.0398, -0.0091)	1.00	1.01
β_{10}- β_{11}	0.0132	0.0131	0.0109	0.0108	1.99E-04	2.03E-04	(-0.0079, 0.0348)	(-0.0084, 0.0348)	1.00	1.00
β_{10}/β_{11}	0.5384	0.5098	0.9448	0.5744	0.0179	0.0107	(-0.0486, 1.532)	(-0.1411, 1.69)	1.00	1.05
σ_b^2	1.19	1.186	0.2042	0.2084	3.64E-03	3.86E-03	(0.8668,1.662)	(0.8563, 1.666)	1.00	1.00
σ_θ^2	0.4021	-	0.0364	-	2.36E-04	-	(0.3359,0.4772)	-	1.00	1.00
α	-	2.482	-	0.2109	-	-	0.0043	(2.093, 2.93)	1.00	1.00
DIC	4868.32	4838.73								

8.7 Epilepsy Data: Posterior summary statistics for the the additive and multiplicative models.

Par.	Mean		SD		MC error		95% Credible interval		$\widehat{R}$	
	Add	Mult	Add	Mult	Add	Mult	Add	Mult	Add	Mult
β_{00}	0.4696	0.6485	0.1874	0.1854	0.0048	0.0042	$(0.0895, 0.8379)$	$(0.2813, 0.9949)$	1.00	1.00
β_{01}	0.736	0.9117	0.1808	0.1835	0.0045	0.0042	(0.3755,1.094)	(0.5538,1.269)	1.00	1.00
β_{10}	-0.0133	-0.0117	0.0076	0.0075	1.42E-04	1.52E-04	(-0.0284, 0.0012)	(-0.0262, 0.0036)	1.00	1.01
β_{11}	-0.0273	-0.0249	0.0078	0.0078	1.38E-04	1.53E-04	(-0.0430, -0.0119)	(-0.0398, -0.0091)	1.00	1.01
β_{10}- β_{11}	0.0132	0.0131	0.0109	0.0108	1.99E-04	2.03E-04	(-0.0079, 0.0348)	(-0.0084, 0.0348)	1.00	1.00
β_{10}/β_{11}	0.5384	0.5098	0.9448	0.5744	0.0179	0.0107	(-0.0486, 1.532)	(-0.1411, 1.69)	1.00	1.05
σ_b^2	1.19	1.186	0.2042	0.2084	3.64E-03	3.86E-03	(0.8668,1.662)	(0.8563, 1.666)	1.00	1.00
σ_θ^2	0.4021		0.0364		2.36E-04		- (0.3359,0.4772)	-	1.00	1.00
α		- 2.482		0.2109		-	- 0.0043	(2.093, 2.93)	1.00	1.00
DIC	4868.32	4838.73								

8.8 Epilepsy Data: Posterior summary statistics for the the additive and multiplicative models.

Par.	Mean		SD		MC error		95% Credible interval		$\widehat{R}$	
	Add	Mult	Add	Mult	Add	Mult	Add	Mult	Add	Mult
β_{00}	0.4696	0.6485	0.1874	0.1854	0.0048	0.0042	$(0.0895, 0.8379)$	$(0.2813, 0.9949)$	1.00	1.00
β_{01}	0.736	0.9117	0.1808	0.1835	0.0045	0.0042	(0.3755,1.094)	(0.5538,1.269)	1.00	1.00
β_{10}	-0.0133	-0.0117	0.0076	0.0075	1.42E-04	1.52E-04	(-0.0284, 0.0012)	(-0.0262, 0.0036)	1.00	1.01
β_{11}	-0.0273	-0.0249	0.0078	0.0078	1.38E-04	1.53E-04	(-0.0430, -0.0119)	(-0.0398, -0.0091)	1.00	1.01
β_{10}- β_{11}	0.0132	0.0131	0.0109	0.0108	1.99E-04	2.03E-04	(-0.0079, 0.0348)	(-0.0084, 0.0348)	1.00	1.00
β_{10}/β_{11}	0.5384	0.5098	0.9448	0.5744	0.0179	0.0107	(-0.0486, 1.532)	(-0.1411, 1.69)	1.00	1.05
σ_b^2	1.19	1.186	0.2042	0.2084	3.64E-03	3.86E-03	(0.8668,1.662)	(0.8563, 1.666)	1.00	1.00
σ_θ^2	0.4021	-	0.0364	-	2.36E-04	-	(0.3359,0.4772)	-	1.00	1.00
α	-	2.482	-	0.2109	-	-	0.0043	(2.093, 2.93)	1.00	1.00
DIC	4868.32	4838.73								

8.9 Epilepsy Data: Model Fit

		Estimate		95% Credible interval	
Effect	**Par.**	Add.	Mult.	Add.	Mult.
Intercept placebo	β_{00}	0.470	0.649	(0.090,0.838)	(0.281,0.995)
Intercept treated	β_{01}	0.736	0.912	(0.376,1.094)	(0.554,1.269)
Slope placebo	β_{10}	-0.013	-0.012	(-0.028, 0.001)	(-0.026, 0.004)
Slope treated	β_{11}	-0.027	-0.025	(-0.043, -0.012)	(-0.040, -0.009)
Slope difference	$\beta_{10} - \beta_{11}$	0.013	0.013	(-0.008, 0.035)	(-0.008, 0.035)
Slope ratio	β_{10}/β_{11}	0.538	0.510	(-0.049, 1.531)	(-0.141, 1.692)
Variance rand. int.	σ_b^2	1.193	1.186	(0.867,1.662)	(0.856, 1.666)
Variance overdispersion	σ_θ^2	0.402	–	(0.336,0.477)	–
Multiplicative overdisp.	α	–	2.482	–	(2.093, 2.931)
DIC		4868.32	4838.73		